

Arduino Traffic Light Project (Without Resistor)

■ Important Note

This project is for demonstration purposes only. Using LEDs without resistors may damage LEDs or Arduino pins if used for long durations.

Components Required

- 1 Arduino Uno
- 2 Breadboard
- 3 3 LEDs (Red, Yellow, Green)
- 4 Jumper Wires

LED Leg Identification

- Long leg = Positive (Anode)
- Short leg = Negative (Cathode)

Connections (No Resistor)

- 1 Red LED long leg → Arduino Pin 8
- 2 Yellow LED long leg → Arduino Pin 9
- 3 Green LED long leg → Arduino Pin 10
- 4 All LED short legs → Arduino GND

Arduino Code

```
int redLED = 8; int yellowLED = 9; int greenLED = 10; void setup() { pinMode(redLED, OUTPUT);  
pinMode(yellowLED, OUTPUT); pinMode(greenLED, OUTPUT); } void loop() { digitalWrite(redLED,  
HIGH); digitalWrite(yellowLED, LOW); digitalWrite(greenLED, LOW); delay(3000);  
digitalWrite(redLED, LOW); digitalWrite(yellowLED, HIGH); digitalWrite(greenLED, LOW);  
delay(1000); digitalWrite(redLED, LOW); digitalWrite(yellowLED, LOW); digitalWrite(greenLED,  
HIGH); delay(3000); }
```